

PROOF OF CONCEPT

Vitsika

Malagasy ant identification from a single photograph

vitsika is Malagasy for ant. Genus-level, 27 genera, built on public data and a frozen public model.

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VitsikaMalagasy ant genus identification · proof of concept

IdentifyGeneraDistributionAtlasMethods

← Identify another specimen

QUERY IMAGE



casent0002219_p.jpg · embedded in 1240 ms

GENUS PREDICTION

Royidris 100%
subfamily myrmicinae

✓ Held-out test specimen casent0002219 (AntWeb label: *Royidris notorthotenes*) · correct

1.	<i>Royidris</i> myrmicinae	99.7%
2.	<i>Nesomyrmex</i> myrmicinae	0.2%
3.	<i>Pheidole</i> myrmicinae	0.1%

See where this lands on the atlas →Where *Royidris* has been collected →

← Identify another specimen

hf-hub:imageomics/bloclip-2 · probe 2026-08-30 · probabilities calibrated on cross-validated training folds (temperature scaling, 5 folds); still an answer among the 27 genera only.

Most similar training specimens

cosine similarity of BloCLIP 2 embeddings; species labels are AntWeb's, not the model's



Royidris notorthotenes
myrmicinae
casent0002259 0.950
© April Nobile · [AntWeb](#)



Royidris robertsoni
myrmicinae
casent0077935 0.981
© April Nobile · [AntWeb](#)



Royidris notorthotenes
myrmicinae
casent0002257 0.889
© April Nobile · [AntWeb](#)



Monomorium bifidoclypeatum
myrmicinae
casent0173589 0.861
© April Nobile · [AntWeb](#)



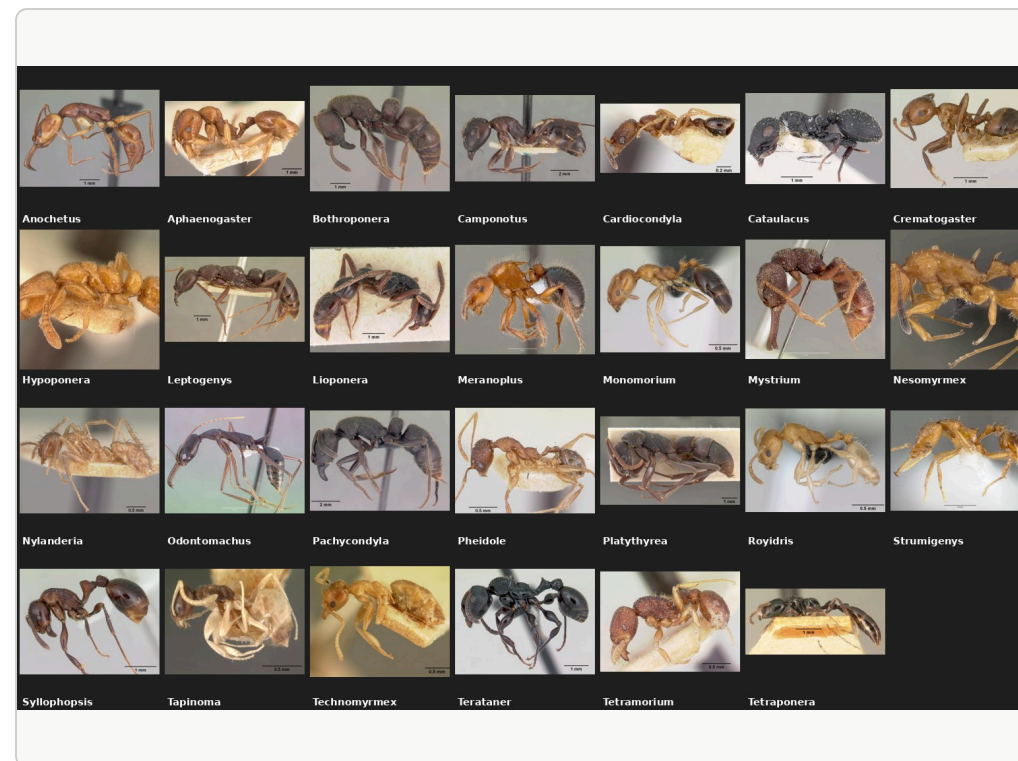
Nesomyrmex flavus
myrmicinae
casent0189276 0.858
© Erin Prado · [AntWeb](#)

Vitsika: *vitsika* is Malagasy for ant. Images © their photographers via [AntWeb](#) (CC BY-SA); metadata via GBIF (CC BY).

[Methods & limitations](#)

Genus-level identification is scarce expertise - a local first pass multiplies it

- AntWeb's GBIF dataset held **141,430** Malagasy ant records on 2026-08-26, **6,995** with images, across **60** imaged genera (GBIF occurrence API; doi:10.15468/wqmqjtt). Naming even the genus takes a specialist and a microscope.
- Three decades of collecting, imaging and determination by the California Academy of Sciences and Malagasy field teams make this data trainable at all. MBC's Phase III roadmap: 'with training, more steps can be done in Madagascar' (madagascarbio.org/monitoring.html).
- Vitsika demonstrates the division of labour: a photo-to-genus first pass runs in under a second on an ordinary computer in Antananarivo - fast enough to process a full night's camera-trap haul locally, with standard optimizations still to apply. It clears the routine identifications and flags what it cannot place, so specialist time goes to the ambiguous and the possibly new.



One AntWeb profile shot per genus in scope (27). Sources: reports/01_explore.txt, README.md.

From GBIF to 1,236 clean images in 27 genera

141,430

AntWeb records for Madagascar on GBIF (doi:10.15468/wqmjjt); 6,995 with an image



4,354

specimens, 39 genera: workers, profile view, one image each, ≥ 10 per genus



1,297

images obtained (29.8 %): 1,160 Wikimedia Commons + 137 GBIF cache



1,236

images, **27 genera**, 8 subfamilies; 986 train / 250 held-out test

Finding 1: GBIF's image cache is empty for AntWeb

4,215 of 4,354 cache requests returned 404 (96.8 %), uniformly across years and record age. antweb.org blocks scripts (Cloudflare) and datacenter IPs (403), including GBIF's own fetcher. The working source is the 2008-2014 bulk upload to Wikimedia Commons (33,081 files), matched by specimen code.

Finding 2: caste is not in GBIF's interpreted fields

GBIF maps sex to Male / Female / Other, so every worker arrives as null: 4,540 of 5,857 specimens. The caste has to be read from the verbatim record (dwc:sex): 4,539 workers, 673 queens, 640 males, and only workers are kept.

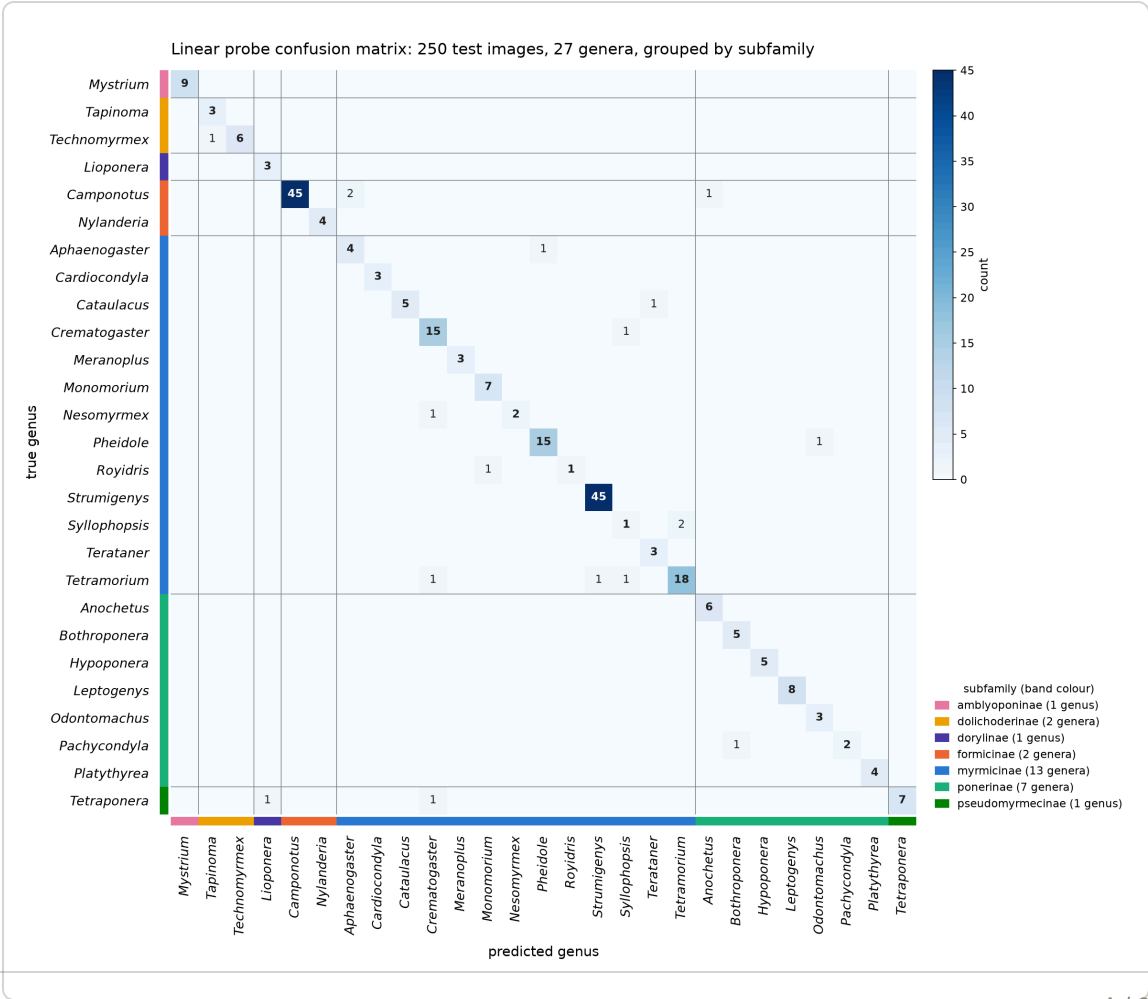
RESULTS

Linear probe on frozen BioCLIP 2 embeddings: 92.8 % top-1 on 250 held-out specimens

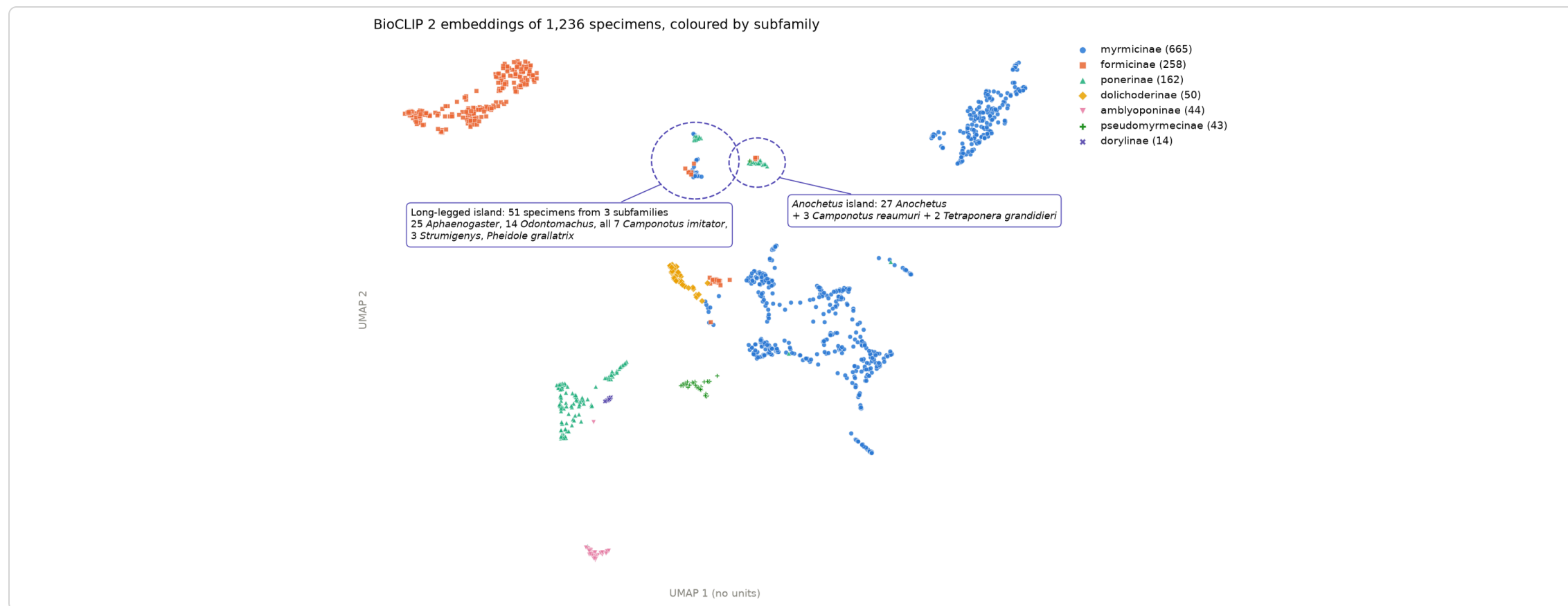
METHOD	TOP - 1	TOP - 3	MACRO - F1
Zero-shot, "a photo of {genus}, a genus of ant"	74.4 %	94.0 %	0.590
Linear probe (deployed)	92.8 %	98.4 %	0.885
Nearest neighbour, cosine	91.6 %	93.2 %	0.834

986 train / 250 test, 27 genera, split grouped by specimen, stratified by genus (seed 42). Bare genus name as the zero-shot prompt: 32.8 % top-1. Embeddings: imageomics/bioclclip-2 ViT-L/14, 768-d, no fine-tuning.

18 errors: 12 genuine look-alikes or species unseen in training (*Sylophopsis* vs *Tetramorium*, *Royidris* vs *Monomorium*), 3 poor photos, 2 odd angles, 1 unexplained. Almost every confusion stays inside a subfamily block.



The embedding space recapitulates taxonomy, with body-plan islands



Subfamilies form clean clusters; where the model "errs", it groups *Camponotus imitator* with long-legged *Aphaenogaster* and *Odontomachus*, a resemblance a taxonomist would recognise, not noise.

Three things this codebase already supports

1

Complete the image set

70.2 % of the manifest's images are missing. `05_download_antweb.py` + `dataset_full.csv` are ready for a residential IP or an AntWeb bulk export; re-thresholding brings *Vitsika*, *Tanipone*, *Carebara* and 9 more genera into scope (39 in total).

2

Species-grouped split and an open-set reject

Group the split by species (one flag in `03_build_dataset.py`) and turn the nearest-neighbour similarity the API already returns into a "none of the 27" reject (probability calibration: done, live since 2026-08-30).

3

Multi-view, multi-caste embeddings

The harvest already holds 6,634 head and 5,127 dorsal media rows plus 673 queens and 640 males; `config.yaml` (`views_priority`, `castes`) and `06_embed.py` make view-wise embedding and late fusion a configuration change.